Dark Energy from the Secluded Majorana SIDM Lagrangian

Summary of the `dark-energy-T-breaking` module

Omer Pesach
ORCID: 0009-0004-8178-8131

March 28, 2026

1 Context: Paper 1 — Secluded Majorana SIDM

The companion paper [1] establishes a *secluded* Majorana dark matter model with a light scalar mediator ϕ:

$$\mathcal{L}_{\rm SIDM} = \tfrac{1}{2}\bar{\chi}(i\,\partial\!\!\!/ - m_\chi)\chi + \tfrac{1}{2}(\partial\phi)^2 - V_0(\phi) - \tfrac{1}{2}\bar{\chi}\big(y_s + iy_p\gamma^5\big)\chi\,\phi \tag{1}$$

with three free parameters (m_χ, m_ϕ, α_s) determining self-interactions. The Variable Phase Method (VPM) solver, Majorana spin statistics ($w_\ell^{\rm even} = 1$, $w_\ell^{\rm odd} = 3$), and Boltzmann relic density are all implemented and validated. Key results:

- 80,142 viable points from a 2.8M-point scan; 17 relic-constrained benchmarks with $\Omega_\chi h^2 = 0.120$ and $\chi^2/\nu < 0.68$ for 13 astrophysical systems.
- MAP benchmark: $m_\chi = 98.19\,\text{GeV}$, $m_\phi = 9.66\,\text{MeV}$, $\alpha_D = 3.274 \times 10^{-3}$.
- Majorana fingerprint: $\sigma_T^{\rm Maj}/\sigma_T^{\rm Dir} = 2.2$ at $v \approx 60$ km/s (triplet dominance 82%).
- Fully secluded: $\sigma_{\rm SI} = \sigma_{\rm SD} = 0$ (Higgs portal excluded by 3×10^4).

Published data and code: Zenodo DOI `10.5281/zenodo.19225823`.

2 Extension: Dark Energy from the Same Lagrangian

2.1 Core idea

We promote the CP phase θ in the Yukawa coupling to a dynamical field σ (a dark axion / dark pion):

$$y_s + iy_p\gamma^5 \;\longrightarrow\; y\,e^{i\gamma^5\sigma/f} \;=\; y\cos\!\Big(\frac{\sigma}{f}\Big) + iy\sin\!\Big(\frac{\sigma}{f}\Big)\gamma^5 \tag{2}$$

where f is the decay constant, and σ acquires a potential from dark QCD chiral symmetry breaking:

$$\boxed{V(\sigma) = \Lambda_d^4\Big(1 - \cos\frac{\sigma}{f}\Big)} \tag{3}$$

with Λ_d the dark QCD confinement scale and $m_\sigma = \Lambda_d^2/f$ the dark pion mass.

2.2 Dark QCD sector

A confining dark gauge group $\mathrm{SU}(2)_d$ with 3 Majorana quarks produces the dark pion σ. The confinement scale is set by dimensional transmutation:

$$\Lambda_d = \mu \cdot \exp\!\Big(-\frac{2\pi}{b_0\,\alpha_d(\mu)}\Big), \qquad b_0 = \tfrac{19}{3}\ (\mathrm{SU}(2)_d,\ 3\ \text{Majorana}) \tag{4}$$

With $\alpha_d(m_\chi) \approx 0.032$ and $\mu = m_\chi = 98\,\text{GeV}$, this yields $\Lambda_d \approx 2\,\text{meV}$ — the dark energy scale — *without* fine-tuning. The coupling $\alpha_d \sim 0.03$ is comparable to α_{EM}.

2.3 Cosmological misalignment

At reheating, σ sits at a misalignment angle $\theta_i \equiv \sigma_i/f \sim 3$ rad (near the CP-preserving hilltop at $\theta = \pi$). While $H \gg m_\sigma$, the field is frozen by Hubble friction. Today $V(\sigma_{\text{today}}) \approx \rho_\Lambda$.

3 The Coupled ODE: $\sigma(t) \otimes$Friedmann $\to H_0$

We solve a coupled ODE system in $N = \ln a$ (e-fold time):

$$\frac{d\sigma}{dN} = p, \tag{5}$$

$$\frac{dp}{dN} = -(3-\varepsilon)\,p - \frac{V'(\sigma)}{H^2}, \tag{6}$$

$$H^2 = \frac{\rho_r + \rho_m + V(\sigma)}{3M_{\text{Pl}}^2 - \frac{1}{2}p^2}, \tag{7}$$

$$\varepsilon = \frac{\frac{4}{3}\rho_r + \rho_m + H^2 p^2}{2M_{\text{Pl}}^2 H^2} \tag{8}$$

where $\rho_r \propto a^{-4}$, $\rho_m = \rho_\chi(a) + \rho_b(a) \propto a^{-3}$.

Inputs (6 Lagrangian parameters):

$$\underbrace{m_\chi,\ m_\phi,\ \alpha_D}_{\text{SIDM sector} \to \Omega_\chi h^2}\ , \qquad \underbrace{f,\ \Lambda_d,\ \theta_i}_{\text{dark QCD} \to V(\sigma)}$$

Output: H_0 [km/s/Mpc] — derived, not input.

4 Key Results (Tests 23–34)

4.1 H_0 as a prediction

With $\Lambda_d = 2\,\text{meV}$, $f = 0.27\,M_{\text{Pl}}$:

θ_i [rad]	H_0 [km/s/Mpc]	Comparison
3.0 (near hilltop)	~ 71	Between Planck and SH0ES
3.1	~ 73	SH0ES-compatible
2.887 (DESI fit)	~ 66.5	Planck-compatible
π (exact hilltop)	~ 67.4	$\equiv$ CDM limit

4.2 DESI DR1 match (Test 27)

The benchmark $\theta_i = 2.887$ rad reproduces the DESI DR1 equation-of-state:

$$w_0 = -0.727 \quad (\text{DESI mean: } -0.727 \pm 0.067) \tag{9}$$

Our model *naturally predicts quintessence* ($w > -1$), distinguishing it from ΛCDM.

4.3 Dark energy is transient (Test 32)

Unlike ΛCDM, the dark energy in our model *dies*: σ eventually oscillates around $V = 0$.

- Acceleration stops at $a \approx 1.57$ (~ 5 Gyr from now)
- $\langle w \rangle \to 0$ after oscillations begin (pressureless matter)
- No Big Rip, no heat death — eternal decelerated expansion

4.4 Sensitivity analysis (Test 33)

Parameter	$\partial H_0/\partial x$	Role
θ_i	$+12.1\,\mathrm{km/s/Mpc/rad}$	Primary dial (initial condition)
$\ln \Lambda_d$	$+90.7\,\mathrm{km/s/Mpc/e\text{-}fold}$	Amplitude ($V \propto \Lambda_d^4$)
$\ln f$	$+3.0\,\mathrm{km/s/Mpc/e\text{-}fold}$	Mass scale (barely matters)

Hubble tension $= \Delta\theta_i \approx 0.47$ rad. *Not fine-tuning*: θ_i is a continuous initial condition set by inflation.

4.5 $\Lambda_d \sim$ meV from transmutation (Test 34)

The 122-order hierarchy $M_{\mathrm{Pl}}^4/\rho_\Lambda \sim 10^{122}$ is traded for:

1. A moderate gauge coupling $\alpha_d \sim 0.03$ (comparable to α_{EM})
2. A standard hierarchy $m_\chi/M_{\mathrm{Pl}} \sim 10^{-16}$

Both are unremarkable in particle physics.

4.6 Additional checks

- $\Delta N_{\mathrm{eff}} \approx 0$ (Test 30): natural couplings give $\Gamma/H = 10^{-143}$ — σ never thermalizes. Full thermalization would give $\Delta N_{\mathrm{eff}} = 3.23$ (excluded by Planck); validates secluded nature.
- **2-loop β-function** (Test 31): $b_1 = 33.08$, shifts α_d by -9%. Physics unchanged; amplification factor $= 1002$.
- **f from self-consistency** (Test 29): f is not free once θ_i is known. At DESI $\theta_i = 2.9$: $f \approx 0.17\,M_{\mathrm{Pl}}$ (satisfies the Weak Gravity Conjecture $f < M_{\mathrm{Pl}}$).

5 Model summary

Free parameters	
m_χ, m_ϕ, α_D	Fixed by Paper 1 SIDM + relic
α_d (dark gauge coupling)	$\to \Lambda_d$ via transmutation
θ_i (misalignment angle)	Set by inflation (cosmological IC)
f	Derived from $V(\theta_i) = \rho_\Lambda$ (not free)
H_0	**Output** of the coupled ODE

The model adds **2 genuinely new parameters** (α_d, θ_i) to the 3-parameter SIDM model, and *derives* H_0 and the dark energy equation of state $w(z)$.

6 Comparison with ΛCDM

Property	ΛCDM	This model
w today	-1 (exact)	≈ -0.95 (quintessence)
w future	-1 (always)	$\to 0$ (oscillates)
DE fate	Eternal	**Transient** (dies)
Acceleration	Eternal	Transient (~ 5 Gyr more)
Origin of DE	Bare Λ (fine-tuned 10^{-122})	Transmutation ($\alpha_d \sim 0.03$)
Falsifiable by $w \neq -1$?	No	**Yes** (DESI, Euclid)
Parameter count for DE	1 (Λ)	2 (α_d, θ_i)

7 Code and data

All code is available in the `dark-energy-T-breaking/` submodule:

- `hunt_H0/layer8_cosmic_ode.py` — coupled σ–Friedmann ODE solver
- `hunt_H0/rg_transmutation.py` — Λ_d from α_d via RG
- `hunt_H0/desi_comparison.py` — DESI DR1 $w(z)$ match
- `hunt_H0/sensitivity_H0.py` — Jacobian $\partial H_0/\partial(\text{params})$
- `hunt_H0/future_universe.py` — cosmic fate / transient DE
- Research journal: 34 tests documenting the full discovery path

Paper 1 code and data: https://zenodo.org/records/19225823
Layer 8 code and data: Zenodo DOI `10.5281/zenodo.19274637`

References

[1] O. Pesach, *Secluded Majorana SIDM: Constraints and Predictions*, Zenodo (2026). DOI: 10.5281/zenodo.19225823.